etc, and produce methane gas. This gas is normally used for cooking. However, if we have to use it for CNG vehicles or for generators, the raw biogas has to be treated; it contains carbon-dioxide, water vapour, and hydrogen sulphide gas. When these unwanted gases are removed, we get 90% concentration of methane, which is then compressed. Compressed biogas is a good substitute for CNG.

If biogas is used to generate electricity using standard available bio-gas generators:

- One cow provides 1000kWh (units) of electrical energy per year
- One cow provides 2.75kWh energy per day
- Hundred cows provide 275kWh energy per day

If biogas is used for making bio-CNG:

- One cow provides 1.5 cubic metres of raw biogas per day
- Hundred cows provide 1500 cubic metres of raw biogas per day
- If purified, 100 cows provide 900 cubic metres of pure methane per day

Agri-voltaic farms

Agri-voltaics is a combination of agriculture and solar photovoltaic (PV) energy generation. PV power plants need large stretches of land. Usually, the land selected for PV installations is barren land. In many places fertile land is also being used. If agriculture land has to be used, then agri-voltaics should be preferred.

In agri-voltaics, the PV panels are installed at a height of more than 244cm (8 feet). And only 50% area is covered by the panels, as shown in Fig. 1. Fig. 2 shows close up view of an assembly of three panels each.

The gaps between the panels allow the sunlight to fall on the land below, permitting cultivation. In the shade of solar panels, various types of vegetable and flowers can be easily grown. As the land is partially shaded, the plants need less water. In spite of less water usage the yield is the same or sometimes better.

The presence of plants provides cool air to the solar panels. Cooler PV panels produce increased power output. The water used for cleaning the panels irrigates the plants and does not go waste. The farmer is assured of regular income by selling electric power generated by the PV panels. Also, he can do the farming and double his income.

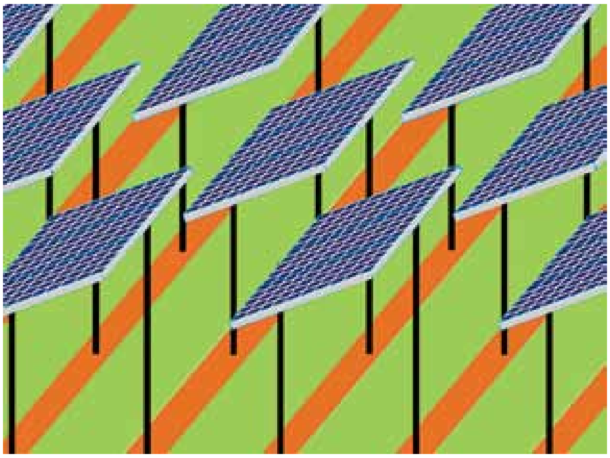

Fig. 2: Close up view of Agri-voltaics farm showing 3 panels attached together and mounted in alternate slots in each row

Let us have a look at a PV system design:

1. One acre area = 43560sq ft
2. Area of each 300Wp PV panel = 6.5 × 3.25 = 21.125sq ft
3. Number of 300Wp panels possible = 43560/21.125 = 2062
4. Number of panels if only 50% area is covered = 1031
5. Panels after leaving 13% area for construction = 900
6. With 3 panels in each PV assembly, number of assemblies = 300
7. Slots with 20 × 30 arrangement = 600 (adjacent slots left vacant)
8. Power output = 300Wp × 900 = 270,000Wp or 270kWp
9. Charging power (assuming 90% efficiency of chargers) = 270 × 0.9 = 243kWp
10. Average charging power available for 8 hrs on a sunny day = 200kW
11. Maximum number of 25kW charges which could be run = 8
12. Total energy generated in 8 hours = 200kW × 8hrs = 1600kWh or 1600 units

Battery based energy storage systems

Electric grid gets power from various sources, such as coal power plants, hydel generators, nuclear, solar, and wind. As a greater number of solar and wind power plants are getting installed, the renewable energy content in the mix is increasing.

Unfortunately, the renewable energy sources are highly variable and unpredictable. This poses problem for the grid stability. Sometimes there is excess power generation and other times there is sudden reduction in generated power.

In order to stabilise the grid, there is need for short-term energy storage. A battery bank can be used in such a situation. When there is excess energy, the batteries can store the energy. When there is demand for energy from the grid, stored energy from the batteries can be used.

Even though use of batteries appears to be a simple solution, the cost of these batteries is very high. Luckily, this being a land based application, size and weight of batteries is not very critical. Hence, even the cheaper lead-acid batteries can be used.

Another option for reducing the cost is to use discarded lithium-ion batteries from electric vehicles. Such batteries still have about 75% storage capacity. These batteries can be used for a few years, as their second life.

Gravity batteries. As the demand